web-ccoedemo-dev-pytho

Flask implementation of the `web-ccoedem` Microsoft Entra authentication demo.

Current Scope

- Demonstrates MSAL and Easy Auth flows in one Python web app
- Uses `app.py`, `templates/`, and `static/` as the main app surface
- Deploys through both Azure DevOps pipelines and GitHub Actions in this repo
- Includes additional architecture, pipeline, and deployment notes in `docs/`

Key Files

- `app.py`
- `requirements.txt`
- `templates/`
- `static/`
- `.github/workflows/azure-webapp.yml`
- `azure-pipelines.yml`
- `run_from_package.yml`
- `docs/`

Documentation

- `docs/MOC.md`: map of content for the full documentation set
- `docs/ARCHITECTURE.md`: app structure, auth modes, runtime behavior, and deployment architecture
- `docs/PIPELINES.md`: GitHub Actions plus both Azure DevOps pipeline paths in this repo
- `docs/DEPLOYMENT_METHODS.md`: side-by-side comparison of supported deployment styles
- `docs/VALIDATION.md`: repository scan and validation notes

Use `docs/MOC.md` as the best starting point when you want the full document map.

Deployment

Four deployment methods are documented for this app:

- GitHub Actions ZIP deploy with App Service build automation
- Azure DevOps pipeline ZIP deploy with Oryx build
- App Service source control sync from Azure DevOps Git
- App Service Run From Package

See `docs/DEPLOYMENT_METHODS.md` for the full comparison and operational details for each method.

For the repo-managed CI/CD flows specifically, see `docs/PIPELINES.md`.

Notes

- The GitHub Actions workflow is aligned with the sibling Node.js and .NET stacks for stage naming, manual deploy gating, deployment prechecks, and publish prechecks.
- The deploy stage supports primary, secondary, and third Linux App Service targets.
- Secondary and third targets are optional and are skipped safely when blank or not found.
- The build stage resolves the runner's local Python 3.12 interpreter directly and deploys to Linux App Service Python runtimes.
- Deployments have been validated successfully against Linux App Service Python 3.12 runtime stacks.
- The application code supports 32-bit Python interpreters when dependencies are installed for that 32-bit target.
- The current Azure deployment path remains Linux App Service Python, which is not a selectable 32-bit built-in App Service stack.
- This repo is application-focused and does not have a root Terraform stack.
- Pre-commit is enabled with lightweight file hygiene checks for local commits.